EE2004: Digital Signal Processing

Tutorial 7

(March 17, 2026)

1. Consider the filter $H(z)$ with difference equation $y[n] - y[n-2] = x[n] - x[n-6]$.
   (a) Determine the poles and zeros of $H(z)$. Is $H(z)$ a causal and stable filter?
   (b) Derive the magnitude response of $H(z)$, and sketch it for $0 \leq \omega \leq \pi$.

2. Let
$$H(z) = \frac{1}{1 - 2r\,\cos\theta\; z^{-1} + r^2\, z^{-2}}$$
   (a) Show that the peak of $|H(e^{j\omega})|$ occurs at $\omega_0 = \cos^{-1}\left(\frac{1+r^2}{2r}\,\cos\theta\right)$.
   (b) Calculate the theoretical peak location when $r = 0.8$ and $\theta = \pi/10$ and verify it using MATLAB.

3. Consider the causal and stable LTI filter with transfer function $H(z) = \dfrac{1}{1 - a\,z^{-1}}$, where $0 < a < 1$. Find the closed-form expression for the frequency at which the magnitude-squared transfer function is half the peak value.

4. Consider the improved bandpass filter with zeros at $z = \pm 1$:
$$H_{\text{BP}}(z) = \frac{1-\alpha}{2} \cdot \frac{1 - z^{-2}}{1 - \beta(1+\alpha)z^{-1} + \alpha z^{-2}}$$
   The poles are at $re^{\pm j\theta}$, and hence $r = \sqrt{\alpha}$ and $2\sqrt{\alpha}\cos\theta = \beta(1+\alpha)$.
   (a) Show that the magnitude-squared response can be written as
$$\frac{(1-r^2)^2\,\sin^2\omega}{(1+r^2)^2\,(\cos\omega - \beta)^2 + (1-r^2)^2\,\sin^2\omega}$$
   (b) Observe $|H_{\text{BP}}(e^{j\omega})|^2 = K\; y^2/(x^2+y^2)$ where x, y are functions of ω and K is a constant. Show that the *maximum* occurs when $x = 0$. Hence determine the value of ω in terms of r and θ at which this maximum is attained.

5. **Computer assignment** Plot the unwrapped phase angle for a stable all-pass filter. Verify its magnitude response is unity for all frequencies. Try various orders and pole locations (once you choose the poles, the zeros are automatically fixed). Does the phase function exhibit any particular characteristic? What are the values of the phase at 0, π, and 2π? Is the phase angle at $\omega = 2\pi$ related to the filter order?

6. **Computer assignment** Compute and plot the phase response of $H(z) = [(1 + z^{-1})/2]^6$. (a) Determine analytically $\angle H(e^{j\omega})$ and using this expression plot the phase response. (b) Use MATLAB's `freqz` command and obtain the phase response and compare it with the results obtained from the previous plot.

7. **Computer assignment** Consider the notch filter

$$H(z) = \frac{1 - 2\cos\theta\, z^{-1} + z^{-2}}{1 - 2r\cos\theta\, z^{-1} + r^2 z^{-2}}$$

Let $\theta = \pi/3$. Plot the magnitude frequency response for $r = 0.9$, $r = 0.95$, $r = 0.99$, and $r = 0.999$. ◇